Deepfake Detection Pipeline using EfficientNet-B0 and Robustness Analysis

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Abstract

Deepfake technology leverages advanced deep learning architectures to synthesize highly realistic manipulated media, posing significant threats to information integrity, identity security, and financial systems. While state-of-the-art detectors achieve exceptional accuracy on clean benchmark datasets, their performance typically degrades when subjected to real-world distortions such as social media compression and noise. This report presents a comprehensive binary deepfake detection pipeline utilizing an ImageNet-pretrained EfficientNet-B0 model fine-tuned on the Celeb-DF dataset. Individual frames are extracted and preprocessed using Multi-task Cascaded Convolutional Networks (MTCNN) for precise facial localization and cropping. The fine-tuned model demonstrates strong classification capabilities, achieving a frame-level Area Under the Receiver Operating Characteristic Curve (AUC) of 0.9588 on the original test set. Furthermore, we conduct a systematic robustness analysis under five common real-world perturbations: Gaussian Blur, JPEG Compression, White Gaussian Noise, Colour Contrast, and Colour Saturation. The results indicate that while the architecture remains stable under compression and color modifications, it experiences significant performance degradation under blur and additive noise, underscoring the vital need for robust forensic tools in practical applications.

1. Introduction

1.1 Background

Deepfake technology uses deep learning to generate highly realistic manipulated images and videos. Although it has applications in entertainment and digital media, it can also be misused for spreading misinformation, identity theft, and financial fraud. As deepfakes become increasingly realistic, reliable automated detection methods are essential.

1.2 Motivation

Most existing deepfake detection methods achieve high performance when evaluated on clean benchmark datasets. However, in practical applications, media files are rarely available in their original form. Images and videos shared through social media platforms often undergo various transformations such as compression, blurring, noise addition, and colour modifications. These perturbations can significantly degrade the performance of detection algorithms, making them less reliable in real-world scenarios.

1.3 Objective

The primary objectives of this project are:

- To develop a binary deepfake detection pipeline using the Celeb-DF benchmark dataset.
- To preprocess video data through frame extraction and face detection using MTCNN.
- To fine-tune an ImageNet-pretrained EfficientNet-B0 model for real versus fake face classification.
- To evaluate model performance using the Frame-level Area Under the Receiver Operating Characteristic Curve (AUC).
- To analyse the robustness of the trained model under Gaussian Blur, JPEG Compression, White Gaussian Noise, Colour Contrast, and Colour Saturation perturbations.

1.4 Scope of the Project

This project implements a practical deepfake detection pipeline using EfficientNet-B0 as the underlying classification model. The implementation includes dataset preprocessing, face extraction, model training, and robustness evaluation. The project follows the evaluation methodology proposed in the benchmark paper by using Frame-level AUC and perturbation-based robustness analysis. Unlike the benchmark study, which compares multiple forensic detection algorithms, this implementation focuses on a single detector, EfficientNet-B0. Therefore, the primary emphasis is on reproducing the benchmark evaluation methodology rather than performing cross-model comparisons. The implementation is limited to image-level deepfake detection using extracted facial frames from the Celeb-DF dataset.

2. Literature Review

Automated deepfake detection has evolved rapidly alongside improvements in generative models, particularly Generative Adversarial Networks (GANs) and diffusion models. Early forensic techniques focused on hand-crafted biological features, such as eye-blinking frequencies, head pose inconsistencies, and irregular blood flow patterns. However, as synthesis technology improved, these specific anomalies became less pronounced.

Modern deepfake detection architectures primarily leverage deep Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs) to learn subtle artifact patterns directly from data. Architectures such as XceptionNet and EfficientNet variations have established benchmarks in frame-level classification due to their capability to extract dense spatial features. Nonetheless, as highlighted by contemporary literature, a prominent gap remains in cross-dataset generalization and robustness against environmental distortions. Social media networks heavily compress and alter uploaded media files, which effectively strips away the high-frequency pixel artifacts that deep learning backbones heavily rely on. This review emphasizes the critical balance between native dataset accuracy and generalizable robustness across diverse digital channels.

3. Dataset and Preprocessing

3.1 Celeb-DF Dataset

The Celeb-DF dataset was selected for this project as it contains high-quality real and manipulated celebrity videos, making it a widely used benchmark for deepfake detection. Since EfficientNet-B0 is an image-based model, videos were converted into facial images before training.

3.2 Dataset Preparation

The complete dataset was divided into training, validation, and testing subsets to ensure unbiased model evaluation. The training set was used for learning model parameters, the validation set was used for hyperparameter tuning and model selection, and the testing set was reserved for final performance evaluation.

The preprocessing pipeline follows the sequence shown below:

Video → Frame Extraction → Face Detection → Image Preprocessing → Train/Validation/Test Dataset

This pipeline converts raw videos into cropped facial images that are subsequently used for training the EfficientNet-B0 model.

3.3 Frame Extraction

Since deepfake manipulation is performed on facial regions within videos, individual video frames were extracted before model training. Videos were processed frame by frame using OpenCV. Instead of processing complete videos directly, representative frames were extracted to reduce computational complexity while preserving sufficient facial information. Each extracted frame was stored as an RGB image for further processing. This conversion transforms the video classification problem into an image classification problem, allowing the use of EfficientNet-B0 for binary classification.

3.4 Face Detection using MTCNN

Not all regions of a video frame contribute equally to deepfake detection. Most forgery artifacts are concentrated around the face. Therefore, Multi-task Cascaded Convolutional Networks (MTCNN) were employed to automatically detect and crop facial regions from each extracted frame. MTCNN performs three sequential tasks:

- Face detection
- Facial landmark localization
- Face alignment

The detected facial region was cropped from the original frame while removing unnecessary background information. This preprocessing step improves training efficiency by allowing the neural network to focus only on the manipulated facial features rather than irrelevant image regions.

3.5 Image Preprocessing

After face detection, each cropped facial image underwent preprocessing before being used for training. The preprocessing pipeline included:

- Resizing all images to 224 × 224 pixels, which matches the input size of EfficientNet-B0.
- Conversion of images into PyTorch tensors.
- Pixel value normalization using the ImageNet mean and standard deviation, ensuring compatibility with the pretrained EfficientNet-B0 weights.

3.6 Data Augmentation

Deep learning models often suffer from overfitting when trained on limited or highly correlated data. To increase the diversity of the training dataset, data augmentation techniques were employed during model training. The augmentation pipeline included random horizontal flipping, random rotations, and other geometric transformations supported by the PyTorch framework.

The augmentation process generated multiple transformed versions of the same image by applying random transformations such as horizontal flipping and small rotations. These augmentations enabled the model to learn more robust facial representations and reduced its dependence on specific orientations or viewpoints. During testing, no random augmentation was applied. Only deterministic preprocessing operations such as resizing and normalization were performed to ensure a fair evaluation of model performance.

4. Proposed Methodology

4.1 Overview

The proposed deepfake detection pipeline consists of video frame extraction, face detection, image preprocessing, model training, and performance evaluation. Videos from the Celeb-DF dataset are first converted into image frames. Facial regions are then detected using MTCNN, pre-processed, and classified as Real or Fake using an EfficientNet-B0 model. Finally, the trained model is evaluated using Frame-level AUC and perturbation-based robustness analysis.

Video → Frame Extraction → MTCNN Face Detection → Image Preprocessing → EfficientNet-B0 → Real/Fake Prediction

4.2 Frame Extraction and Face Detection

Individual frames were extracted from each video using OpenCV. Since deepfake artifacts are primarily present in facial regions, MTCNN was used to detect and crop faces from each frame. The cropped facial images were resized to 224 × 224 pixels before being used for training.

4.3 Image Preprocessing

The extracted face images were converted into PyTorch tensors and normalized using the ImageNet mean and standard deviation. During training, data augmentation techniques including random horizontal flipping, colour jitter, Gaussian blur, and Gaussian noise were applied to improve the model's robustness and generalization capability. For validation and testing, only resizing and normalization were performed.

4.4 Model Training

An ImageNet-pretrained EfficientNet-B0 model was fine-tuned for binary classification. The final classification layer was modified to produce two output classes corresponding to Real and Fake images. The model was trained using the Cross-Entropy Loss function and the Adam optimizer.

4.5 Performance Evaluation

Model performance was evaluated using the Frame-level Area Under the Receiver Operating Characteristic Curve (AUC), as recommended in the benchmark paper. In addition to the original test set, robustness was analysed by evaluating the trained model under five image perturbations: Gaussian Blur, JPEG Compression, White Gaussian Noise, Colour Contrast, and Colour Saturation.

5. Experimental Setup and Evaluation

5.1 Experimental Setup

The proposed model was implemented using Python and the PyTorch deep learning framework. An ImageNet-pretrained EfficientNet-B0 model was fine-tuned for binary classification on the Celeb-DF dataset. The Adam optimizer and Cross-Entropy Loss function were used for model training. Training was performed using mini-batch gradient descent, while the validation set was used to select the best-performing model. The major training parameters used in this work are summarized in Table 5.1.

Table 5.1: Training Configuration

Parameter	Value
Model	EfficientNet-B0
Input Size	224 × 224
Optimizer	Adam
Loss Function	Cross-Entropy Loss
Learning Rate	0.0001
Batch Size	32
Epochs	5

5.2 Evaluation Methodology

Model performance was evaluated using the Frame-level Area Under the Receiver Operating Characteristic Curve (AUC), as recommended in the benchmark paper. AUC measures the model's ability to distinguish between real and fake images across all classification thresholds, making it more reliable than accuracy for binary classification.

To evaluate robustness, the trained model was tested under five image perturbations:

- Gaussian Blur
- JPEG Compression
- White Gaussian Noise
- Colour Contrast
- Colour Saturation

For each perturbation, the Frame-level AUC was computed and compared with the original test set to analyse the robustness of the proposed model under realistic image degradations.

6. Results and Discussion

6.1 Model Performance

The proposed EfficientNet-B0 model was evaluated on the Celeb-DF test dataset using the Frame-level Area Under the Receiver Operating Characteristic Curve (AUC). The model achieved a Frame-level AUC of 0.9588 on the original test set, demonstrating a strong ability to distinguish between real and manipulated facial images.

6.2 Robustness Analysis

To evaluate robustness, the trained model was tested under five common image perturbations. The corresponding Frame-level AUC values are presented in Table 6.1.

Table 6.1: Frame-level AUC under Different Perturbations

Perturbation	Frame Level AUC
Original (Clean Test Set)	0.9573
Gaussian Blur	0.7043

JPEG Compression	0.9160
White Gaussian Noise	0.5213
Colour Contrast	0.8836
Colour Saturation	0.9312

6.3 Discussion

The experimental results indicate that EfficientNet-B0 performs well on the original Celeb-DF dataset. The model maintained relatively high AUC under JPEG Compression, Colour Contrast, and Colour Saturation, indicating good robustness to these perturbations.

However, the performance decreased significantly under Gaussian Blur and White Gaussian Noise. These perturbations remove or distort important facial features required for deepfake detection, making classification more challenging. Overall, the proposed model demonstrates effective deepfake detection capability while maintaining reasonable robustness under several common real-world image perturbations.

7. Conclusion and Future Work

7.1 Conclusion

This project presented a deepfake detection pipeline based on EfficientNet-B0 using the Celeb-DF benchmark dataset. The preprocessing pipeline included video frame extraction, face detection using MTCNN, image preprocessing, and transfer learning for binary classification. The trained model was evaluated using the Frame-level Area Under the Receiver Operating Characteristic Curve (AUC), following the evaluation methodology proposed in the benchmark paper.

The model achieved a Frame-level AUC of 0.9588 on the original test set. Robustness analysis under different image perturbations showed that the model remained relatively stable under JPEG Compression, Colour Contrast, and Colour Saturation, while Gaussian Blur and White Gaussian Noise resulted in noticeable performance degradation. The experimental results demonstrate that EfficientNet-B0 is an effective backbone for image-based deepfake detection and that perturbation-based evaluation provides a more comprehensive assessment of model robustness than accuracy alone.

7.2 Future Work

The following improvements can be considered in future work:

- Evaluate additional deepfake detection architectures such as XceptionNet and Vision Transformers.
- Extend the implementation to video-level deepfake detection instead of frame-level classification.
- Perform cross-dataset evaluation to study model generalization.
- Improve robustness against severe image perturbations using advanced data augmentation and adversarial training techniques.
- Explore lightweight models suitable for real-time deployment on edge devices.